3.1.2 Divisions of Surveying

Surveying is broadly classified into two primary divisions.

a. Plane surveying

b. Geodetic surveying

a. Plane surveying

When the extent of area surveyed is comparatively small, then the effect of curvature is not considered as the surface of the earth is assumed to be flat or plane is called plane surveying.

b. Geodetic surveying

When the extent of area surveyed is large or degree of accuracy required is more, then the effect of curvature of earth is taken into consideration and the survey is called geodetic surveying. It is also known as trigonometrical survey. In our country it is carried out by the great trigonometrical department of India.

3.1.3 Classification of Surveying

The detailed classification of surveying is shown in the a flow chart below.

Classification of surveying Based on

Nature of field	Object of surveying	Method of surveying	Instruments used
1. Land	1. Engineering	1. Triangulation	1. Chain
2. Marine (or) Navigation	2. Archaelogical	2. Traverse	2. Compass
3. Astronomical	3. Geological		3. Plane table
4. Space	4. Military		4. Level
	5. Mine		5. Theodolite
			6. Tacheometry
			7. Aerial and photography

3.1.4 Uses of Surveying

Some of the uses of surveying from engineer's point of view are as follows

1. To know the relative position and shapes of the area to be surveyed (hills, valleys, lakes, rivers, etc.)
2. To establish boundary lines of property.
3. To find the areas of boundary, catchment, ayacut, etc.
4. To calculate the volumes of earthwork in cutting and embankment, capacity of reservoir, etc.
5. To select suitable alignment for engineering woks (roads, railway, canal, bridge, etc.) and to locate the same properly.
6. To determine the suitability of a site for any engineering work
7. To collect data for planning , analysis, design, cost estimation, etc.

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